

Wireless and Mobile Security (CY315)

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Chapter 5 – How Do WLANs Work?

- Wireless & Mobile Security
 - Understanding WLAN Foundations Before WLAN Security

Why Understanding WLAN Operation Matters

- Understand wireless communication fundamentals
 - Learn WLAN topologies and 802.11 standards
 - Recognize RF behaviors and limitations
 - Security begins with operational understanding

Learning Outcomes

- Describe WLAN topologies (BSS, ESS, IBSS, MBSS)
- Differentiate between 802.11 standards
- Explain frequency spreading concepts
- Describe AP functionality
- Distinguish AP vs Wireless Bridge
- Understand antenna types

Fundamental Requirement of WLAN

- Devices must operate on a common RF frequency
 - Must comply with regulatory standards
 - Operate in ISM unlicensed spectrum

What is ISM Spectrum?

- ISM = Industrial, Scientific, Medical band
- Unlicensed wireless spectrum
- Used for:
 - Wi-Fi
 - Bluetooth
 - RFID
 - Zigbee
 - IoT devices
- Defined by:
 - International Telecommunication Union (ITU)
 - Regulated locally by agencies like Federal Communications Commission (FCC)
- ⚠ Even though unlicensed, devices must follow strict technical rules.

Key ISM Bands for WLAN

Frequency	Bandwidth	Availability
2.4 GHz	100 MHz	Worldwide
5 GHz	150 MHz	Worldwide
6 GHz	1200 MHz	Recently Approved
60 GHz	High	Short-range only

Why 2.4 GHz is Popular

- **Advantages:**

- Better wall penetration
- Longer range
- Sufficient bandwidth
- Used by Wi-Fi, Bluetooth, IoT

- **Disadvantages:**

- Congested
- Interference from microwaves
- Shared by many technologies

Why 5 GHz is Preferred in Enterprise

- Higher throughput
 - Less interference
 - More channels
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- Shorter range than 2.4 GHz
 - More attenuation through walls

Introduction of 6 GHz (Wi-Fi 6E)

- 1200 MHz additional spectrum
 - Less congestion
 - Supports multiple non-overlapping channels



Security Note:

More spectrum leading to less congestion and fewer collision-based exploits.

WLAN Anatomy – Station (STA)

- Any 802.11 capable device
 - Laptop, smartphone, IoT device
 - Access Point is also a STA

Access Point (AP)

- **Role:**
 - Central communication hub
 - Connects wireless clients to wired LAN
 - Acts as Layer 2 bridge
- **AP acts as:**
 - Layer 2 bridge
 - RF controller
 - Traffic mediator

Client Requirements

- To act as Wi-Fi client, device must have:
 - ✓ Radio card (TX/RX)
 - ✓ Antenna
 - ✓ 802.11 protocol support
- Important:
Wi-Fi is **Half-Duplex**
- Only one device transmits at a time.

Duplex Modes Comparison

Mode	Example	Behavior
Simplex	FM Radio	One-way only
Half-Duplex	Wi-Fi	One at a time
Full-Duplex	Ethernet	Simultaneous TX/RX

Client Association Process

- **Client must:**
 - Detect AP
 - Associate
 - Authenticate
 - Connect
- **Roaming occurs when:**
 - Stronger AP signal detected
 - Same SSID
 - Different channel

SSID (Service Set Identifier)

- SSID = Network name
- For seamless roaming:
 - ✓ Same SSID
 - ✓ Same security configuration
 - ✓ Different channels
 - ✓ 15–20% coverage overlap

Passive Scanning

- Client listens for:
Beacon frames
 - AP continuously broadcasts:
 - SSID
 - Capabilities
 - Security type
 - Supported data rates
- Client connects to strongest signal.

Active Scanning

- Client sends:
 - Probe Request
- Types:
 - Directed Probe (specific SSID)
 - Null Probe (blank SSID)
- AP responds with:
 - Probe Response

Security Risk of Active Scanning

- Active scanning can expose device to:
- **Rogue Access Points**
- Attacker:
 - Sets powerful AP
 - Same SSID
 - Captures traffic
- Common attack:
Evil Twin Attack

802.11 Service Sets Overview

- **Defined topologies:**
 1. Basic Service Set (BSS)
 2. Extended Service Set (ESS)
 3. Independent BSS (IBSS)
 4. Mesh BSS (MBSS)

Basic Service Set (BSS)

- Single AP with multiple clients
 - All communication via AP
 - Used in homes and small offices

Extended Service Set (ESS)

- Multiple APs connected via Ethernet
 - Supports seamless roaming
 - Requires coverage overlap
- **Types:**
 - Seamless roaming (overlapping coverage)
 - Nomadic roaming (no overlap)
 - Collocation (100% overlap)

Independent BSS (IBSS)

- Ad hoc mode
 - Peer-to-peer communication
 - No central access point

Mesh BSS (MBSS)

- Mesh topology
 - Client-to-client and AP-to-AP communication
 - Used in large-scale deployments

Evolution of 802.11 Standards

- Originally defined by:
 - Institute of Electrical and Electronics Engineers (IEEE)
- Letter-based naming:
 - 802.11a
 - 802.11b
 - 802.11n
 - 802.11ax

Wi-Fi Alliance Naming

Wi-Fi Name	IEEE Standard	Year
Wi-Fi 1	802.11b	1999
Wi-Fi 2	802.11a	1999
Wi-Fi 3	802.11g	2003
Wi-Fi 4	802.11n	2009
Wi-Fi 5	802.11ac	2014
Wi-Fi 6	802.11ax	2019

Managed by:
Wi-Fi Alliance

Wi-Fi 4 (802.11n)

- Up to 600 Mbps
 - Dual-band support
 - Introduced MIMO

Wi-Fi 5 (802.11ac)

- 5 GHz only
 - Wider channels (80/160 MHz)
 - Introduced MU-MIMO

Wi-Fi 6 (802.11ax)

- Designed for high-density environments
 - Up to 10–12 Gbps theoretical speed
 - Uses OFDMA and 1024-QAM

Why Wi-Fi 6 Was Needed

- Older Wi-Fi uses CSMA/CA
 - Congestion in high-density areas
 - Wi-Fi 6 improves efficiency and capacity

OFDMA in Wi-Fi 6

- Divides spectrum into smaller subchannels
 - Multiple users transmit simultaneously
 - Improves spectrum efficiency

MU-MIMO Improvements

- Bidirectional in Wi-Fi 6
 - Supports multiple simultaneous streams
 - Uses beamforming

60 GHz (802.11ad)

- Up to 7 Gbps
 - Short range
 - Line-of-sight required

Security Relevance

- Rogue AP attacks
 - Evil twin attacks
 - RF jamming and congestion exploits